

MUHAMMAD HUSSAIN

Senior Full Stack Developer • AI Integration Specialist • 4+ Years Experience

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PROFESSIONAL SUMMARY

Self-driven **Senior Full Stack Developer & AI Integration Specialist** with 4+ years of hands-on experience independently designing, building, and deploying scalable web applications, SaaS platforms, and AI-powered solutions from the ground up. Deep expertise in React.js, Next.js, Node.js, and TypeScript, combined with practical knowledge of LLM integration — including RAG pipelines, MCP servers, Claude/OpenAI/Gemini APIs, and prompt engineering. Proven ability to embed intelligent AI features (chatbots, semantic search, AI assistants) into production web applications, backed by strong cloud and DevOps skills (AWS, Docker, CI/CD). Autonomous professional who owns the full project lifecycle — from architecture and development to deployment and optimization.

WORK EXPERIENCE

Full Stack Developer & AI Integration Engineer

Jan 2025 – Present

Studio Cog — Remote

- Architect and deliver cloud-native SaaS applications using TypeScript, React 18, and Node.js with AI-powered features
- Integrated LLM APIs (Claude, OpenAI GPT-4) and built RAG pipelines using vector embeddings to power intelligent search and chatbots
- Built MCP (Model Context Protocol) servers to connect LLMs with application data sources and external APIs
- Design and optimize PostgreSQL schemas and implement secure REST APIs with JWT authentication
- Deploy and manage applications on AWS (EC2, S3, Lambda) with CI/CD pipelines for reliable continuous delivery

Full Stack Engineer & AI Solutions Developer (Freelance)

Jan 2024 – Jan 2025

Upwork — Remote

- Delivered 10+ custom SaaS applications for international clients, maintaining a 5-star satisfaction rating
- Integrated AI capabilities — LLM chatbots, Gemini API, Codex-powered code assistants — into client web platforms
- Implemented RAG (Retrieval-Augmented Generation) systems for document Q&A; and knowledge-base applications
- Integrated Stripe payments and third-party AI APIs enabling platforms processing 10K+ monthly transactions
- Reduced client app load times by up to 40% through code splitting, lazy loading, and server-side caching strategies

Senior Frontend Developer

Mar 2023 – Jan 2024

PIFD — On-site

- Independently developed and maintained responsive web platforms using React.js and TypeScript
- Translated complex Figma UI/UX designs into pixel-perfect, production-ready React components
- Improved Google Lighthouse performance scores significantly via asset optimization and rendering improvements
- Took full ownership of frontend codebase — ensuring clean, maintainable, and scalable code architecture

Frontend Developer

Dec 2022 – Mar 2023

Intellisoft Solutions — On-site

- Built dynamic, responsive UIs with React.js and Redux across 5+ client projects
- Refactored legacy JavaScript to ES6+, reducing bundle size by 20% and improving load times by 25%
- Ensured full cross-browser compatibility (Chrome, Firefox, Safari, Edge) and mobile responsiveness

- Built responsive user interfaces using HTML5, CSS3, and JavaScript under senior developer guidance
- Converted Figma/Zeplin designs into accurate, functional, and responsive web pages
- Optimized websites for mobile performance and improved cross-browser compatibility
- Rapidly grew from fundamental skills to independently handling frontend tasks within the first year

AI & LLM INTEGRATION EXPERTISE

Specialized in integrating cutting-edge AI/LLM technologies into production web applications and SaaS platforms:

LLM Models:	Claude (Anthropic), GPT-4o (OpenAI), Gemini 1.5 Pro (Google), Codex, Llama (local inference)
AI Techniques:	RAG (Retrieval-Augmented Generation), CAG (Cache-Augmented Generation), Prompt Engineering, Fine-tuning basics, Embeddings & Semantic Search
AI Protocols:	MCP — Model Context Protocol (server & client), Function/Tool Calling, Streaming Responses, Multi-turn Conversations
AI SDKs & APIs:	Anthropic SDK, OpenAI SDK, Google Gemini SDK, LangChain, LlamaIndex, Vercel AI SDK
AI Tooling:	Claude Code (Anthropic), Cursor AI, GitHub Copilot, Gemini CLI, Codex CLI, ChatGPT Advanced, v0.dev
AI in Web Apps:	AI Chatbots & Assistants, Semantic Search, Document Q&A, AI-powered dashboards, LLM-driven automation

TECHNICAL SKILLS

Frontend:	React.js, Next.js, TypeScript, Redux, Tailwind CSS, HTML5, SCSS, Material UI, Framer Motion
Backend:	Node.js, Express.js, Python (Basic), MongoDB, PostgreSQL, RESTful APIs, GraphQL, WebSockets
Cloud & DevOps:	AWS (EC2, S3, Lambda, RDS), Docker, CI/CD Pipelines, Vercel, Git/GitHub, Postman, Nginx

EDUCATION & CONTINUOUS LEARNING

Full Stack Development & AI Engineering — Self-Taught Professional

Sep 2021 – Present

Independent Study, Structured Training & Hands-on Project-Based Learning

- Mastered Advanced MERN Stack, TypeScript, REST/GraphQL API design, and cloud architecture through self-directed study
- Deep-dived into LLM fundamentals: RAG pipelines, MCP protocol, vector databases, embeddings, and AI API integration
- Built production AI-integrated apps: LLM chatbots, document Q&A systems, AI-powered SaaS dashboards
- Continuously upskilling in AI engineering (LangChain, LlamaIndex), DevOps best practices, and AI-assisted development